

- 1 In addition to the superficial logical meaning, logical operations actually have a "masking" meaning. How can the two operations of AND and OR be used as masks? (10%)

**Sol:**

- (1) If you perform an "and 0" operation on a part of a binary number, then no matter what its original value is, it will be cleared to 0 (clear / zero out).
- (2) After some bits are performed with the operation of 'or 1', these bits will definitely be set to 1.

- 2 For the original input words, (1) logical right (15%), (2) arithmetic right (15%), (3), and cyclical right. (15%)

The original input word      1010 1101 0011 0111 0000 0101 0011 1110

**Sol:**

- (1) logical shift :

The original input word

=>1010 1101 0011 0111 0000 0101 0011 1110

Perform logical shifts to the right

=>0000 1010 1101 0011 0111 0000 0101 0011

- (2) arithmetic shift :

The original input word

1010 1101 0011 0111 0000 0101 0011 1110

Perform arithmetic shifts to the right

1111 1010 1101 0011 0111 0000 0101 0011

- (3) circular shift:

The original input word

1010 1101 0011 0111 0000 0101 0011 1110

Perform a loop to move right

1110 1010 1101 0011 0111 0000 0101 0011

3. Below are three examples with detailed step-by-step animations

Example 1: MUL R1, R2, R3 (Arithmetic Operation) (15%)

Multiply contents of R2 (4) and R3 (5), store in R1

Example 2: LOAD R1, [800] (Memory Load) (15%)

[800] =152

Example 3: BRANCH IF ZERO R1, label (15%)

If R1 == 1

Sol